5 Additional objects and experimental design

The square demonstrated one equilibrium solution. We now compare geometry, mesh size, loading and supports to test how widely its behaviour applies.

5.1 Objects and shared physical settings

We retain the unit square as a control and add two objects (Fig. 1). The longer rectangle tests the effect of increasing the distance between support and load. The perforated square tests removing material while keeping the outer boundary fixed. We compare both deformation and solver effort. Both changes also alter material area, so neither isolates shape at constant area.

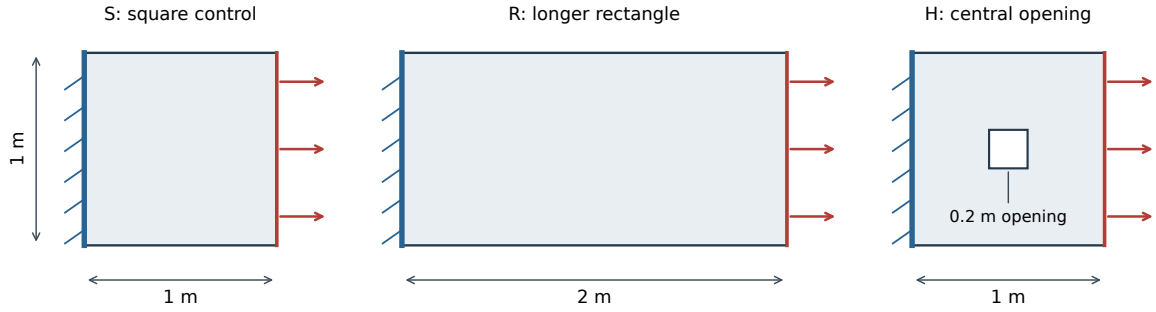


Figure 1: Geometries at a common scale, in metres, not meshes or deformations. Blue edges are clamped in both directions; arrows indicate uniform rightward traction on red edges. Other edges are free.

Coordinates are in metres. The square's outer boundary is $(0,0),(1,0),(1,1),(0,1)$; the rectangle's is $(0,0),(2,0),(2,1),(0,1)$. The perforated square retains the first boundary and adds the opening $(0.4,0.4),(0.4,0.6),(0.6,0.6),(0.6,0.4)$. Outer vertices run counterclockwise, hole vertices clockwise, and each list closes back to its first vertex. The opening is empty space, not a softer material.

All baseline cases retain Section 4's plane-stress settings: $E = 200$ GPa, $\nu = 0.3$, $t = 0.01$ m, no body force and zero displacement in both directions along the whole left edge. Uniform rightward traction $p = 100$ MPa acts on the whole right edge. Each edge is one metre high, giving the same total force $F_0 = 10^6$ N. Other boundaries are traction-free. Material and thickness stay fixed throughout.

5.2 Meshes that preserve the physical case

We use Triangle's boundary-constrained meshing through its Python interface, as introduced in Section 2 [1]. All three objects, including the square control, will use this route; Section 4's structured mesh is therefore a separate demonstration, not a matched-mesh geometry comparison. We request a minimum triangle angle of 25° and maximum triangle areas of 0.005, 0.00125 and 0.0003125 m^2. These are area limits, not guaranteed edge lengths or error bounds. Equal area limits need not give equal numbers of unknowns.

Boundary segments, holes and load/support endpoints must be preserved. Edge forces are integrated as in Section 2, not assigned equally per node, so refinement preserves the physical loading. Before solving, we check positive triangle areas, boundary coverage, an empty opening and one connected solid region. Both support choices below fix at least two distinct points in both directions, preventing translations and rotation. We check the supported matrix and record element counts, free unknowns and mesh quality. The accompanying notebook records the boundary splits, hole seed and mesher options, and stores the generated meshes.

5.3 Controlled comparisons and recorded responses

Table 1 defines the comparisons. Except under refinement, we use the middle area limit. Load and support studies reuse each object's mesh, changing only the named factor from the baseline.

Table 1: Case register, using fixed material and thickness and Section 3's protocol. Results are given in Section 6.

Study	Changed factor	Question and fixed quantities
Geometry	Square, rectangle, perforated square	How do response and solver cost change at fixed traction, clamp and mesh-area limit?
Mesh	Three area limits, on each object	Do displacement measures settle under refinement of the same physical case?
Magnitude	$0.5F_0$, F_0, $1.5F_0$ on each object	Does displacement scale with load when its location, mesh and supports are fixed?
Location	Lower, middle or upper right-edge patch on each object	How does moving the same total force affect deformation at fixed mesh and clamp?
Supports	Full left clamp or central half-clamp on each object	How does releasing part of the boundary affect response at fixed mesh and full-edge load?

For the location study, each patch is 0.25 m high: its vertical interval is $[0, 0.25]$, $[0.375, 0.625]$ or $[0.75, 1]$ m. Each carries F_0 rightward, so its uniform traction is $F_0/(0.25\,\mathrm{m}\,t) = 400$ MPa; the rest of the edge is unloaded. Comparing these three equal-length patches isolates location. Comparing a patch with the full-edge baseline would also change load concentration. The shorter clamp fixes both components on the left-edge interval $[0.25, 0.75]$ m and leaves the rest of that edge traction-free. Its physical length stays fixed under refinement, not its number of nodes.

Physical measures and outlines use the checked same-mesh LU reference. Across meshes, we report absolute successive differences in the edge-length-weighted mean right-edge horizontal displacement and both midpoint components. Differences below demonstrated algebraic uncertainty remain unresolved. Outlines state units and amplification. Solver measurements follow Section 3; conditioning follows Section 2. Fixed-matrix load scaling checks linearity, not directional sensitivity, which needs a separate perturbation test.

Each case must pass Section 4's support and force/moment-balance checks, together with symmetry, positive-definiteness and residual checks; the thresholds are listed in the notebook. Refinement measures mesh dependence, not error against an exact solution. Sharp corners and boundary-condition changes can concentrate stress; peak corner stresses are therefore excluded as convergence measures. Physical interpretation still requires checking the small-strain assumptions.

References

[1] Jonathan Richard Shewchuk, *Triangle: Generating Delaunay triangulations*, Carnegie Mellon University, Triangle documentation, `https://www.cs.cmu.edu/~quake/triangle.delaunay.html`.